





PLATFORM TECHNOLOGIES

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UNIT 1:
Lesson 1 INTRODUCTION TO PLATFORM TECHNOLOGIES

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LEARNING OBJECTIVES

- Define Platform Technologies;
- Identify the types of Platform Technologies
- Explain the importance of Platform Technologies

What is a Platform?

- a group of technologies that are used as a base upon which other applications, processes or technologies are developed.
- In personal computing, a platform is the basic hardware (computer) and software (operating system) on which software applications can be run. This environment constitutes the basic foundation upon which any application or software is supported and/or developed.



Platform Technology

- refers to the specific technological components and tools that make the platform function effectively.
- It encompasses the software, hardware, protocols, and APIs (Application Programming Interfaces) that enable the functionality of a platform.



Platform Technology

- For example, the Amazon Web Services (AWS) platform includes various platform technologies like Amazon EC2 (Elastic Compute Cloud) for computing resources, Amazon S3 (Simple Storage Service) for storage, and AWS Lambda for serverless computing.

BENEFITS OF PLATFORM TECHNOLOGIES

- Increased Innovation- Building a powerful platform lets you cultivate an ecosystem of developers, partners, users, and other collaborators who contribute to – and may drive – innovation at your company.
- Quicker Response Times-Platform companies move faster than their traditional counterparts. When your core products and services frequently change, it forces your employees and your organization to embrace change quickly.
- Reduced Risk-Platforms encourage diversification and thus reduced risk.

TYPES OF TECHNOLOGY PLATFORM

- OPERATING SYSTEMS-Operating systems provide the basic services required to use hardware. These are the lowest level of platform.







TYPES OF TECHNOLOGY PLATFORM

- COMPUTING PLATFORMS- Platforms built on top of operating systems that provide computing functionality in areas such as cloud computing and virtualization. For example, a cloud computing platform that allows you to scale your services by adding and removing virtual machines as required.







TYPES OF TECHNOLOGY PLATFORM

- Database Platforms-Cloud platforms for deploying and managing various types of database such as relational, NoSQL and in-memory databases.





TYPES OF TECHNOLOGY PLATFORM

- Storage Platforms- Platforms for scalable storage of objects and files including APIs and value added services such as resilient storage that is backed up in multiple locations.

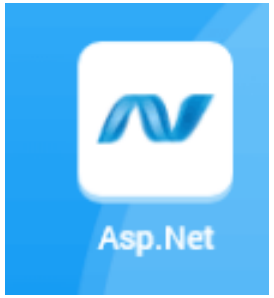




TYPES OF TECHNOLOGY PLATFORM

- Application Platforms - are environments and toolkits for developing and deploying applications, a class of software that is primarily designed to be used by people. For example, a platform that allows developers to build dynamic web applications by executing code and integrating with databases, systems and APIs.





TYPES OF TECHNOLOGY PLATFORM

- Mobile Platforms - include mobile operating systems and environments for building mobile apps.





TYPES OF TECHNOLOGY PLATFORM

- Web Platforms - provide services that are useful to websites and web-based software as a service such as web servers, web application servers, content delivery networks and edge computing.





TYPES OF TECHNOLOGY PLATFORM

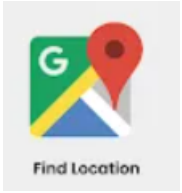
- Content Management Systems- Platform for publishing and managing content, media and documents. These are often used internally in organizations as intranet site and document management system.





TYPES OF TECHNOLOGY PLATFORM

- API Platforms- Cloud platforms for deploying APIs that are typically built around an API gateway that performs functions such as load balancing, latency reduction and rate limiting.





- Analytics- Services for capturing, processing, analyzing and visualizing data. This may include tools for ingesting, processing, querying and managing big data.





TYPES OF TECHNOLOGY PLATFORM

- Security- Security services such as firewalls, identity and access management, directory services, certificates, compliance, reporting, encryption, key management and threat detection.
- Robotics - platform that may include an operating system for robots with a framework for developing and deploying backend systems and services for robots for cloud infrastructure.

TYPES OF TECHNOLOGY PLATFORM

- Internet of Things -platforms may include an operating system for devices and a cloud platform with specialized APIs for internet of things in areas such as device management, IoT security and analytics.
- AI Platforms- Services that are based on AI such as a voice synthesis service and tools for building your own AI such as machine learning API. This may also include environments for running your AI that are optimized for machine learning such as machine learning database.

TYPES OF TECHNOLOGY PLATFORM

- Game Platforms-Environments that are optimized for running game services such as back-ends for mobile games. These may include services such as 3D game engines, AR and VR APIs

